

Aditya Kaleeswar Gurusamy Kumaravel

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EDUCATION

Vellore Institute of Technology (VIT) | Vellore, India

B.Tech., Computer Science and Engineering | Sep 2022–Jul 2026

RESEARCH AND ENGINEERING EXPERIENCE

University of Texas Rio Grande Valley — AIM Lab | Remote

Research Intern | Jan–Jul 2026

- Built a replicate-aware prediction framework for engineered cementitious composites (ECC): grouped 620 tensile tests into 276 unique mixes to prevent composition leakage, then trained target-specific CatBoost quantile models for tensile stress and strain.
- Calibrated 80% prediction intervals with conformal adjustment (80.4% stress and 82.6% strain coverage; group-level R^2 0.749/0.605). Connected the models to support-constrained NSGA-II inverse design, rejecting high-strength/high-ductility targets outside the observed mix domain.
- Developed a rover-image-to-fracture pipeline that segments individual concrete cracks, extracts skeleton arc length and width, and calibrates image geometry to supply boundary conditions for mechanics-based severity scoring.
- Designed a mesh-free PINN with an embedded Williams crack-tip field and a learnable stress-intensity amplitude, constrained by positivity and a near-tip cage loss. Recovered Mode I stress intensity within ~6.5% of an independent finite-element reference across five seeds and tracked analytical geometry factors for $a/W = 0.10$ – 0.30 .

Samsung Research Bangalore — PRISM Program | Remote

Research Intern | Oct 2024–Jun 2025

- Designed BQ2Viz, a multi-agent system for business-question answering and visualization across multiple datasets, combining ChromaDB retrieval, analysis planning, and LangGraph chart generation.
- Added a code-execution and repair loop for generated charts; reached 84.57% accuracy on complex multi-table VisEval queries, above the NVAGENT baseline, with an 11.5% error rate.

Team Vyadh (Mars Rover Team), VIT | Vellore, India

Autonomous Systems Engineer | Apr 2024–Sep 2025

- Architected frontier-based exploration for unknown GPS-denied terrain, selecting reachable frontiers from live occupancy grids and dispatching Nav2 goals in a navigate–re-evaluate loop.
- Co-developed a C++/PCL server classifying RealSense D455 point clouds as pits or obstacles and fused RTAB-Map/OctoMap SLAM for 3D mapping; team placed 4th at IRDC 2025 and 13th at IRC 2025.

PROJECTS

Stress Stack — Repository-to-Evaluation Task Pipeline [GitHub](#)

Python, AST/tree-sitter, Git/GitHub REST, Docker

- Built a pipeline that turns software repositories into reproducible tasks for evaluating coding models and harnesses: it pins dependencies, containerizes tests, maps code and coverage, and packages historical problems with a pre-fix tree, reference fix, verifier, and execution evidence.
- Implemented issue-first mining of closed issues and their merged PR fixes, recovering the original problem statement and true pre-fix commit; a language-spec-driven AST/tree-sitter graph resolves symbols, imports, containment, and inheritance to map the functions affected by each task.

CRAFT — Autonomous Chemo-Visual Hazard Mapping [GitHub](#)

ROS 2 Humble, Nav2, SLAM Toolbox, CUSUM, Jetson Orin Nano

- Built a rover framework that covers and maps unknown GPS-denied spaces while locating gas leaks and visual hazards: a rapidly covering graph samples the discovered frontier, CUSUM gas anomalies trigger local fine sweeps, and fused gas and visual detections form a risk map that identifies hazard type and location and guides subsequent coverage.
- Evaluated in GADEN, a ROS-compatible gas-dispersion simulator with known source locations, and on a Jetson rover: mean source-localization error was 0.54 m in a multi-source setting and 0.53 m in a maze; high-concentration hazards were detected within the first 2% of mission time. Visual inference ran at 24.3 ms/frame with 0.845 mAP@0.5.

Alpha Stack — Terminal-Native Coding-Agent Harness [GitHub](#) | Undergraduate capstone

Python, C++/Tree-sitter, Go/Bubble Tea, Docker, multi-model APIs

- Built a terminal-based coding harness with a Go/Bubble Tea interface that accepts a natural-language project request and returns a generated, build-tested codebase with a trace of planning, tool use, and repairs.
- Architected a requirements-to-blueprint-to-file-contract pipeline with parallel generation and dependency-graph context; a planner/corrector loop runs builds and tests, diagnoses failures, and revises the affected files.
- Designed a 40-task benchmark and reproducible evaluation environment for coding models and agentic harnesses on end-to-end project development in CUDA, Go, Rust, and TypeScript; tracked build/test results, cost and failure telemetry, external acceptance checks, and an independent LLM-as-judge examiner verdict.

PATENTS

A Platform-Based System and Method for Force-Responsive Train State Detection

Published Indian patent application no. 202641007500

An Adaptive Multi-Sensor Edge Computing Node for Autonomous Laboratory Safety Monitoring

Published Indian patent application no. 202641058589

TECHNICAL SKILLS

Languages and data: Python, C/C++, Go, JavaScript/TypeScript, SQL, PostgreSQL

ML: CUDA, OpenCV, PyTorch, TensorFlow, LangGraph, CrewAI, RAG, vector databases, Harbor, Prime Intellect Verifiers

Robotics: ROS 2 Humble, Nav2, SLAM Toolbox, RTAB-Map, OctoMap, Gazebo, Isaac Sim, Intel RealSense SDK, NVIDIA Jetson

Systems: Docker, Git, Linux, Kubernetes, AWS, tree-sitter